MEDUR RODOSHI AUNGSHU

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EDUCATION

Missouri State University Fall 2025 – Present
M.S. (Ongoing) in Computer Science
Graduate Research Assistant

Khulna University of Engineering & Technology (KUET) April 2024
B.Sc. in Energy Science and Engineering, Khulna, Bangladesh
CGPA: 3.17/4.00 (**3.42** in the last 60 credits)

RESEARCH INTERESTS AND EXPERIENCES

Research Interests: Human Computer Interaction, Human-AI Collaboration.

GPU-CPU Memory Transfer Latency Benchmarking 2025
Evaluated data transfer efficiency between CPU and GPU by testing multiple buffer sizes on a production GPU server. Developed a test setup that repeated each transfer thousands of times in both directions to obtain reliable latency measurements, including best-case and worst-case scenarios. Additionally, compared system performance when idle and under active CPU load to assess the impact of competing workloads on memory transfer latency.

KV Cache Memory Scaling in LLM Inference Systems 2026
Conducted an empirical study investigating how GPU memory usage grows in large language model servers as inputs get longer and more users send requests at the same time. Set up varied workloads across different context lengths and concurrency levels, and tracked KV cache usage live through the server's own metrics to see where memory pressure builds up under real conditions.

Tail Latency Characterization Under Concurrent LLM Inference 2026
Examined at how response times in a large language model server get worse as more users send requests simultaneously, particularly when those requests involve very long inputs. Designed a testing framework to capture latency at different concurrency levels, which showed that as contention increases, the slowest responses get hit disproportionately harder than the average.

Numerical Investigation of Thermal Performance of Buildings Integrated with Phase Change Materials 2023
Undergraduate Thesis
For my undergraduate thesis, I explored how phase change materials, substances that absorb and release large amounts of heat as they melt and solidify, could help keep buildings cooler without relying on active cooling systems. Used DesignBuilder and EnergyPlus to simulate how these materials would perform when embedded into the roofs of single-story buildings across different climate zones in Bangladesh. Measured how much energy could be saved and how significantly indoor thermal comfort improved, giving a clearer picture of how PCMs could be a practical solution for hot climate construction.

PROJECTS

Automated Single Axis Solar Tracking Based Battery Charging System 2022
Built a solar tracking system using an Arduino microcontroller and C++ that automatically adjusts a solar panel's angle to follow the sun throughout the day, rather than staying fixed in one position. Ran a side-by-side comparison between the tracking system and a standard static panel to measure exactly how much extra energy the moving system could capture. Also designed and integrated a battery storage setup to make sure the harvested energy was stored and used as efficiently as possible.

SKILLS

Programming Languages C, C++, Python, Arduino Programming, SQL

HONORS & AWARDS

Merit Scholarships
Received Merit Scholarships for a total of three semesters in the Department of Energy Science and Engineering at KUET, Bangladesh.

IELTS Academic Achievement
Band Score: **7.0** | Listening: **8.0** | Reading: **8.5** | Writing: **6.0** | Speaking: **6.0**
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